# Project Proposal: Is Rainfall Sufficient to Explain Reservoir Levels?

## Introduction and Motivation

Water sustainability and resource management are becoming increasingly critical global challenges. Rapid population growth, climate change, urbanization, and irregular precipitation patterns are placing significant pressure on freshwater resources worldwide. In many regions, droughts and water shortages are expected to become more frequent and severe in the near future. For this reason, understanding the factors that influence reservoir and dam occupancy levels has become both scientifically and socially important.

This project aims to investigate the relationship between weather conditions and reservoir occupancy levels in Istanbul, with a particular focus on whether rainfall alone is sufficient to explain changes in water storage levels. While rainfall is often assumed to be the primary determinant of reservoir capacity, water systems are influenced by many interconnected environmental and temporal factors. Through this project, I aim to explore the extent to which short-term weather conditions can explain fluctuations in dam occupancy and whether additional variables and delayed effects must also be considered.

## Data Sources

The primary dataset used in this project is the "İstanbul Barajlarının Doluluk Oranları" dataset obtained from the İstanbul Metropolitan Municipality Open Data Portal (IBB Açık Veri Portalı). This dataset contains daily occupancy rates of Istanbul dams and spans from the early 2000s to the present day. Since the dataset includes multiple dams and long-term observations, it provides a suitable foundation for time-series analysis.

To complement reservoir data, historical weather information will be collected using public weather APIs such as Open-Meteo. Weather variables will include daily rainfall, snowfall, and temperature measurements for the Istanbul region. These datasets will be merged based on date information to create a unified time-series dataset suitable for statistical analysis and predictive modeling.

## Methodology

The project will begin with data cleaning and preprocessing, including handling missing values and aligning datasets temporally. Exploratory Data Analysis (EDA) will then be conducted to identify distributions, trends, seasonal patterns, and correlations between weather conditions and reservoir occupancy levels.

Following EDA, statistical hypothesis testing will be applied to evaluate whether rainfall and other weather variables have statistically significant effects on dam occupancy. Particular attention will be given to delayed effects by creating lagged and cumulative precipitation variables across different time windows. Seasonal analyses will also be performed to investigate whether reservoir levels vary significantly across seasons.

In addition to statistical analysis, machine learning techniques will be used to model and predict reservoir occupancy levels. Feature engineering methods such as rolling averages, lag variables, and seasonal encoding will be implemented to improve predictive performance. Regression-based models, including Linear Regression and Random Forest Regressor, will be evaluated using metrics such as $R^2$ and Mean Absolute Error (MAE).

## Expected Outcomes

This project is expected to demonstrate that rainfall alone is insufficient to fully explain changes in reservoir occupancy levels. Instead, reservoir systems are likely influenced by cumulative weather patterns, seasonal effects, and delayed environmental processes. The findings may contribute to a better understanding of urban water management systems and highlight the complexity of predicting water availability in large metropolitan areas such as Istanbul.

More broadly, this project reflects the growing importance of sustainable water resource management in the face of climate uncertainty and increasing global water demand.